



UNIVERSITY OF CAPE TOWN
Department of Electrical Engineering
EEE4022F/S - Final Year Project
Graduate Attribute Tracking Form

Student name:	Samson Okuthe	DP Awarded? ☒ Y ☐ N	DP satisfied
Student no:	OKTSAM001	Supervisor name:	Simon Winberg
Date:	September 24, 2026	Date:	September 24, 2026
Student signature:		Supervisor signature:	

Instructions:

Students must explain in this document what they **have already done** and what they **plan to do** to satisfy each Graduate Attribute. Descriptions of each GA is provided at the end of the document. Supervisors respond to the student's plans and current progress, providing additional comments or advice as they see fit. Once the student's progress is deemed sufficient (a few weeks before submission at the due date for this form), supervisors indicate that DP can be awarded.

VERY IMPORTANT: Receiving DP for the course does NOT imply that all GA's have been met in the course. Assessment of GA's only happen in the final marking of the project report.

GA 1: Problem Solving

Student Response:

I formulated the core research question around measurable vehicle dynamic response rather than subjective ride comfort, rejecting a driver-comfort framing in favour of one suited to signal-processing analysis. I root-caused several faults from first principles: a GPIO conflict I resolved via the manufacturer's own schematic; a BLE service-discovery failure I confirmed against an independent reference implementation; and a systematic measurement flaw I caught before any field data existed to be corrupted by it, where my literature review showed that a raw acceleration RMS conflates road roughness with vehicle speed, since the road-to-acceleration transfer function is itself speed dependent. I then conducted a competitive-landscape and literature audit to formulate a specific, evidenced research gap: synchronised multi-sensor vehicle-dynamics characterisation for a three-wheeled cargo vehicle, distinct from existing four-wheel and motorcycle telemetry work and existing commercial telematics-plus-HUD products. The next stage is to run this characterisation on the moving vehicle rather than the bench, since every subsystem validated so far has been proven individually but not yet on the vehicle the research question is actually about.

Supervisor Response:

The student has provided good explanations of the problem-solving tasks for this project, and shows a good understanding tasks to do. The student is following an effective approach to refine the design and to produce a functional prototype. The student shows good understanding of the design process, and shows competence in creative problem solving and design.

GA 4: Investigations, Experiments, and Data Analysis

Student Response:

I conducted hypothesis-test-conclusion investigations throughout: the GNSS field-count assumption, which I hypothesised at 13 fields, tested live, found to be 9, and corrected; the BMS service-UUID assumption, which I tested against a powered pack, found wrong, corrected, and cross-validated against an independent parser; and the sync-pulse timing offset, which I bounded to under one second via a lens-cover test. I designed a controlled experiment for the ride-characterisation feature, using a synthetic sine wave with analytically-known RMS, peak-to-peak, and crest-factor values, validating the computation independently of sensor hardware. A further investigation of mine found two nominally-identical accelerometers disagreeing by about 7% at rest; I hypothesised this as factory tolerance, corrected it via a per-sensor calibration, and confirmed it held under an induced dynamic disturbance too, not only the static case it was derived from. My continuously-run literature pipeline has retrieved 733 candidate sources and confirmed the DOI of 114 of them against the published title. The next stage is to bench-validate the newly sourced VL53L5CX proximity sensor against an independent published characterisation of the same chip before trusting its datasheet figures, and to capture a genuine outdoor GNSS fix to resolve the one remaining untested field-mapping assumption.

Supervisor Response:

The student has developed a good understanding of the investigation, considered hypothesis testing and proceeding with a well-considered and effective approach to the research. As student described clear explanations of the needs and activities involved. Effective planning has been completed and the crucial aspects of effective calibration, and other aspects of experiments and obtaining data.

GA 5: Use of Engineering Tools

Student Response:

I applied, and recognised the limitations of, PlatformIO, Arduino, and FreeRTOS for embedded development; esptool and serial diagnostics for hardware verification; OpenSCAD for a parametric, printable camera-mount design; and Python automation (paramiko, PyMuPDF for schematic verification, PIL for frame analysis) that I built to support testing. I verified tool limitations explicitly rather than assuming them: I did not accept a BMS checksum formula as correct merely because a library accepted it, with no vendor datasheet available to confirm it independently; and I re-checked an OpenSCAD export metric, initially misread as a mount-design fault, against a known-valid control case (a plain reference cube) before accepting or rejecting the concern. The next stage is to complete the LittleFS replay-on-reconnect logic and integrate the VL53L5CX proximity sensor into the existing per-task acquisition architecture once it arrives.

Supervisor Response:

The student has selected a good set of tools, has tested and configured these. The student has formed an effective development environment to proceed with the system development and testing. The student has also recapped, learned and applied relevant knowledge needed to perform relevant experiments and pursue system testing and analysis of the data obtained.

GA 6: Professional and Technical Communication

Student Response:

I have maintained a continuously-updated Methodology chapter documenting my engineering work honestly, including faults found and how I resolved them, and negotiated shared vehicle access in writing with the Controls Lab custodian. I also caught and corrected a cross-chapter inconsistency before it reached a reader: the Introduction described the camera's object detector as an active feature, while my Methodology had documented it as withheld under the ethics constraint. I am now drafting the Results chapter directly from the evidence already logged in Methodology.

Supervisor Response:

The student has attended meetings and shown shown written, provided a draft Introduction and shows good technical documentation and writing competence.

GA 8: Individual Working

Student Response:

I took full individual responsibility for hardware bring-up, firmware debugging, and report authorship, making technical decisions independently, bringing my supervisor specific questions rather than requests for direction, and acting on his guidance. I selected the project's components myself, compared the alternatives against the requirements, and placed the orders; for example, I chose an 8×8-zone time-of-flight sensor over a single-point one to give left, centre and right discrimination across the trike's narrow front. I tested each component on the bench before relying on it, from the two IMUs and their calibration to the battery-management link, modem and camera. I designed and 3D printed a casing for the camera, and located and read the literature that shaped the design and the research gap. I respected shared resources, leaving the inherited dashboard's SD card and another team's vehicle-access slots untouched, and I am now coordinating joint integration onto the vehicle ahead of the field test.

Supervisor Response: The student has predominantly done focused individual work for this project. The student demonstrated good competence in independent work and decision making ability with in regards to the complex design tasks that this project involves.

GA 9: Independent Learning Ability

Student Response:

I repeatedly corrected my own assumptions against primary sources, such as GPIO assignments checked against the schematic and a vehicle-identity error. I built a continuously-running literature pipeline to keep self-correcting rather than trust one search; it surfaced prior art that forced two of my research-gap claims to narrow. A 2026 open-architecture IMU, GNSS and cellular telemetry system for motorcycle dynamics showed that such a unit is not novel in itself, so my first claim now rests on the spatially separated dual IMUs, ride-quality framing and three-wheeler class; and prior work already logs locally alongside a cellular uplink, so my second claim became that uplink reliability is usually taken from specifications rather than measured. A third claim, that camera-based hazard awareness has not been combined with vehicle-dynamics telemetry on auto-rickshaw-class vehicles, survived a targeted search. I also checked the modem against its 652-page AT command manual, which documents sixteen GNSS fields in knots where the fitted modem returns nine, so I treat manuals as hypotheses to test against the hardware.

Supervisor Response: The student has demonstrated good individual working ability and consistency in correcting assumptions that may, after further exploration, be found to be misaligned or inaccurate to main project needs. Thus, the student shows strong ability and dedication to quality work, and this gives clear assurance of having satisfied this GA in addition to a high level of professional work and ability to focus on complex learning and development activities.

GA 1: Problem Solving

Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.

- A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- Conceptually based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

- Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline, much of which is at the forefront of the discipline.

GA 4: Investigations, Experiments and Data Analysis

Demonstrate competence to conduct investigations of complex engineering problems using research methods, including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

The balance of investigation and experiment should be appropriate to the discipline. Research methodology to be applied in research or investigation where the student engages with selected knowledge in the research literature of the discipline.

Note: An investigation differs from a design in that the objective is to produce knowledge and understanding of a phenomenon and a recommended course of action rather than specifying how an artefact could be produced.

GA 5: Use of engineering tools

Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.

- Conceptually based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- Knowledge of engineering practice (technology) in the practice areas in the engineering discipline

A range of techniques, resources and modern engineering and IT tools appropriate to the disciplinary designation of the programme.

GA 6: Professional and Technical Communication

Demonstrate competence to communicate effectively, both orally and in writing, with engineering audiences and the community at large, taking into account cultural, language, and learning differences.

This course evaluates the long report component of this outcome at exit level. Material to be communicated is in an academic or simulated professional context. Audiences range from engineering peers, management and lay persons, using appropriate academic or professional discourse. Written reports (10 000 to 15 000 words plus tables, diagrams and appendices) should cover material at exit-level. Methods of providing information include the conventional methods of the discipline, for example engineering drawings, as well as subject-specific methods.

GA 8: Individual, Team and Multidisciplinary Working

Demonstrate competence to work effectively as an individual, in teams and in multidisciplinary environments. This course evaluates the **individual** working component of this learning outcome at exit level.

Knowledge of professional ethics, responsibilities and norms of engineering practice.

GA 9: Independent Learning Ability

Demonstrate competence to engage in independent learning through well developed learning skills.

Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

Operate independently in complex, ill-defined contexts requiring personal responsibility and initiative, accurately self-evaluate and take responsibility for learning requirements; be aware of social and ethical implications of applying knowledge in particular contexts.

- Openness to constructive feedback, awareness of own limitations, ability to cope with the discomfort of uncertainty and having access to a range of approaches, reflective selfevaluation, curiosity and proactive engagement, resilience, confidence to ask for help and draw from a broad range of stakeholders.
- Reflection of self-learning to begin to recognise if what has been covered meets the needs of the activity or task.